

Math-Stats II

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Question 7e:

```
# Given parameters
n <- 50
p1 <- 1/3
p2 <- 1/3
p3 <- 1/3

# Expected values (mean) of geometric distributions
mu_X <- 1 / p1
mu_Y <- 1 / p2
mu_Z <- 1 / p3

# Variances of geometric distributions
sigma_X2 <- (1 - p1) / p1^2
sigma_Y2 <- (1 - p2) / p2^2
sigma_Z2 <- (1 - p3) / p3^2

# Compute mean and standard deviation of L
E_L <- mu_X + mu_Y + mu_Z
Var_L <- (sigma_X2 / n) + (sigma_Y2 / n) + (sigma_Z2 / n)
SD_L <- sqrt(Var_L)

# Compute P(L > 10) using normal approximation
P_L_10 <- pnorm(10, mean = E_L, sd = SD_L)

# Print the probability
P_L_10
```

```
## [1] 0.9522096
```

Question 15:

(i) Probability that the Sample Variance of Company A Exceeds 0.65 Amp

```
pchisq((0.65 * 9 / 0.04), df = 9, lower.tail = FALSE)
```

```
## [1] 5.269854e-27
```

(ii) Probability that the Sample Standard Deviation of B is 2/3 Times Larger than That of A

```
pf(4/9, df1 = 14, df2 = 9, lower.tail = FALSE)
```

```
## [1] 0.9159394
```

(iv) 75th Percentile of the Distribution of S^{2a}/S^{2b}

```
qf(0.75, df1 = 9, df2 = 14)
```

```
## [1] 1.469714
```

Question 17

```
# Load necessary library
library(stats)

# Given parameters
n <- 5 # Sample size
m <- 3 # Binomial parameter (number of trials)
p <- 0.34 # Probability of success

# Compute the probability mass function (pmf) of  $X \sim \text{Binomial}(3, 0.34)$ 
pmf_values <- dbinom(0:m, size = m, prob = p)

# Compute the probability of  $X_{(3)} \geq 3$ 
# The order statistic  $X_{(3)}$  represents the third largest value in the sample.
# We need to compute  $P(X_{(3)} \geq 3) = 1 - P(X_{(3)} \leq 2)$ 

# Probability that at most 2 values in the sample are below 3
prob_X3_leq_2 <- sum(dbinom(0:2, size = n, prob = sum(pmf_values[1:3])))

# Probability that  $X_{(3)} \geq 3$ 
prob_X3_geq_3 <- 1 - prob_X3_leq_2

# Display the result
cat("P( $X_{(3)} \geq 3$ ) =", round(prob_X3_geq_3, 4), "\n")
```

```
## P( $X_{(3)} \geq 3$ ) = 0.9994
```